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Ecology and diversity of Decapods (Crustaceans) in Rukka Dam, located in Ranchi, Jharkhand.

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Abstract- This study has been conducted to explore the crustacean (decapod) diversity in Jharkhand in an artificially created dam named as Rukka dam, which has remained unexplored for a long period of time. Least study of crustacean has been done as compared to another benthic organism. This study was done on a seasonal basis from November 2022 to October 2023. In the three seasons, the samples were collected during post monsoon (November 2022-Febuary 2023), pre monsoon (March 2023-June 2023), Monsoon (July 2023-October 2023 seasons). Decapod recorded included Prawn, Shrimp, Crab and other aquatic arthropods. Samples were collected from the marked site every season. A total of 14 species of crustacean and decapod have been recorded under 4 families and 5 genera of which 11 species of prawn, 1 species of crab and 2 species of shrimp have been recorded. The most diverse family is Palaemonidae in which 10 species have been recorded. The least number of species were from the Penaeidae and Gecarcinucidae families in which only one type of species was recorded.

Key words: Season, Decapod, Arthropod, Prawn, Palaemonidae

INTRODUCTION

Decapods have been reported to be the most diverse crustaceans. It belongs to the class Malacostraca and phylum Arthropoda. Decapod appears in diverse forms throughout the world.[1] There are around 17500 extant species. Decapods exploits both lotic and lentic regions. Along with this, Decapod are found in all places like ocean, river, lakes, ponds and paddy fields. The majority of crustaceans decapod are marine species.[2] Decapod crustaceans are important members of benthic communities. The feeding habits of decapod are herbivores, detrivores, carnivores and omnivores. Decapods play a major role in the aquatic food web. Decapod represent important food resources.[3] They show symbiotic relationship with other phyla.[4,5] Decapod have a great economic contribution. Some species go through the pelagic larval stage and then becomes adults.[6] It takes them week to month to become adults. These play a significant role in planktonic ecology.

Rukka Dam a very large artificial reservoir of Jharkhand. It is built on the Swarnarekha river. It was opened in 1971. Its catchment area is 717m^2. Many villages are situated on the banks of this reservoir Getalsud, Ramdag, Mahespur, Gandhigram, Childag, Salhanae, Lalgarh, Jhusur, Itma, Baridih, Hatwal, Chakla, Rukka Saweyya and Matatu etc. Drinking water is supplied to the people of Ranchi city from this Reservoir. 500 fisherman families are dependent on this reservoir. 300 fish farming cages have been built in Mahespur village area. This

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reservoir is also the main attraction of tourism. Lack of proper arrangements decreases the crowd of tourists.[1]

Study Area

Two sites in the Hurputoli area of the Rukka reservoir were selected for sample collection The Hurputoli (Haruptoli) village is located in Ormanjhi taluka of Ranchi in Jharkhand, India. The census code of this village is 373656. The total geographical area of Hurputoli (Haruptoli) village is 145.35 Hectares/1.45 km^2. The nearest police station is located in Ormanjhi tahsil.



Fig. 1- Map of Getalsud Dam / Rukka Dam



Fig. 2- Location Map of Hurputoli village

A forest area site was selected and the other site was a pool and near cemetery area. The anthropogenic activity in the forest area was very low in all season. In every season Birds, Sheep's and goats were found in large numbers in the forest area. Whereas at other site of, pool and cemetery areas different activities like boating, fishing etc are done in the pool and cemetery area during pre-monsoon. Whereas in the remaining two seasons, there is constant activity of anthropogenic activity and fishermen are present in every season for catching fish.

MATERIALS & METHODS

- Decapod crustacean were collected seasonally from November 2022 to October 2023.
- To collect crustacean decapods a triangle net (2 mm length and 1 mm width) was manually prepared and used to collect decapod from 4 feet of water.
- Some were handpicked where the water was at a depth of about 2 feet.
- After collecting we fix them in 4% formalin and preserved them in 70% alcohol.
- Identification was done at ZSI, Kolkata.

RESULT & DISCUSSION

The species composition and abundance of each decapod group varied with time and season. This also depends on the limnological characters of the water body. During three season of sample collection, we found a total of 14 species of decapod those who belonged to 4 families and 5 genera. Most of the species recorded were from Palaemonidae family and the least numbers of species were found from the Gecarcinucidae and Penaidae families in which only one species was recorded, 2 species of Atyidae family was also found. Few species were noticed during sample collection in the pre -monsoon season. We recorded large numbers of species during the monsoon season. During the post-monsoon season the availability of species was in good number. During sample collection in every season species were found in more diverse form in forest area sites. Fewer species were recorded in each season at the cemetery and pool area sites because there was always busting and anthropogenic activities there.

Sarade and Patil (2015)[7] recorded 15 species of crustacean decapod from the Dahanu Coastal waters of Palghar district of Maharashtra state belonging to 7 families and 11 genera.

A total 4258 species of crustacean from India were recorded by Devroy (2012)[8]. Pawar (2021)[9] reported 26 species of decapod crustaceans which represent 18 genera and 12 families from mangrove ecosystem of Uran (Raigad) in Navi Mumbai along West Coast in India. Mogalekel *et al.* (2015)[10] recorded 20 species of decapod crustacean from Vembanand lake on the West Coast of India belonging to 5 families and 10 genera. Beleem *et al.* (2014)[11] have recorded 22 species of crabs from the Gulf of Kach Marine National Park under 22 genera belonging to 11 families.

Table 1- Showing Decapod species with classification from Getalsud Reservoir, Ranchi, Jharkhand

Order	Family	Genus	Species
	Penaeidae	*Fenneropenaeus*	*Fenneropenaeus indicus* (Milne Edward, 1837)
	Palaemonidae	*Macrobrachium*	*Macrobrachium malcolmsonii* (Milne Edward, 1844)
			Macrobrachium rosenbergii (De- Man, 1879)
			Peguense (Tiwari, 1952)
			Birmanicum choprai (Tiwari)
			Idea (Heller,1862)
			Mirabili (Kemp, 1917)
Decapoda			*Lamarrei* (Milne Edward, 1837)
		Metapenaeus	*Brevicornis* (Milne Edward, 1837)
			Joyneri (Miers, 1880)
			Affinis (Milne Edward, 1837)
	Gecarcinucidae	*Barytelphura*	*Cunicularis* (Westwood in syker 1836)
	Atyidae	*Caridina*	*Gracilipes* (De- Man 1892)
			Williamsoni jalihal (Shenoy and Sankoli)

Table 2 - Number of genera and species of Decapod families from Getalsud Reservoir, Ranchi, Jharkhand

Sl. No.	Families	No. of Genus	No. of Species
1	Penaeidae	1	1
2	Palaemonidae	2	8
3	Gecarcinucidae	1	1
4	Atyidae	1	2

Table 3 - Distributional pattern of freshwater Decapod of Getalsud Reservoir, Ranchi, Jharkhand

Family	Species	Pre-monsoon		Monsoon		Post-Monsoon	
		Site 1	Site 2	Site 1	Site 2	Site 1	Site 2
Penaeidae	*Fenneropenaeus indicus*	-	+	+	+	+	+
Palaemonidae	*Macrobrachium malcolmsonii*	+	+	+	+	-	+
	Macrobrachium rosenbergii	+	+	+	+	+	+
	Peguense	-	-	-	+	+	+
	Birmanicum choprai	-	-	-	+	-	+
	Idae	-	+	-	+	-	-
	Mirabili	-	-	-	+	-	-
	Lamarrei	-	-	+	+	-	+
	Brevicornis	-	+	-	-	-	+
	Joyneri	-	+	+	+	+	+
	Affinis	-	-	+	+	-	-
Gecarcinucidae	*Cunicularis*	-	-	-	+	-	-
Atyidae	*Gracilipes*	-	+	+	+	+	+
	Williamsoni jalihal	-	+	-	+	+	+

CONCLUSION

This study was done so that crustacean could be explored as much as possible because till now least attention has been given to the study of crustacean in entire Jharkhand. Crustaceans' decapod is an important part of benthic organism of water body. Through their study we can know about the condition of water body because decapod is both tolerance and intolerance to water pollution. Decapod are also helpful in aquatic ecosystem and aquatic food web and by farming and culture of decapod economically support can be obtained and their study record can be very helpful for further studies.

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